

Shahista Tamkeen

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PROFESSIONAL SUMMARY

AI and Machine Learning Engineer with 3+ years of experience architecting autonomous agentic AI systems, enterprise RAG pipelines, and high-throughput production ML models across the BFSI and healthcare sectors. Proficient in Python, Java, PyTorch, LangGraph, AWS/Azure/GCP, PySpark, and Kubernetes to build low-latency inference engines and multi-cloud data pipelines. Proven track record of driving impact by cutting compliance audit cycles from 5 days to 45 minutes and reducing enterprise document turnaround times by 82%.

CORE SKILLS

Programming & Frameworks: Python, SQL, Java, C++, PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, FastAPI, gRPC
Agentic AI & LLM Systems: LangGraph, AutoGen, CrewAI, LangChain, LlamaIndex, AWS Bedrock, GCP Vertex AI, GraphRAG (Neo4j), vLLM, Hugging Face Transformers, PEFT/LoRA, DeepEval, NeMo Guardrails
Vector Search & RAG Architecture: ChromaDB, Pinecone, OpenSearch, pgvector, BGE Embeddings, Cohere Rerank, BM25, Hybrid Search
Machine Learning & Data Science: Statistical Optimization, Time-Series Forecasting (Temporal Fusion Transformer), Named Entity Recognition (LayoutLMv3), Computer Vision (Qwen2-VL, OpenCV), SHAP, SMOTE-NC, Optuna
Data Engineering & ETL: PySpark, Polars, Pandas, Dask, Apache Kafka, Apache Beam, AWS Glue, AWS Lambda, Apache Airflow, Redis, Dask
Cloud & MLOps: AWS (S3, Aurora PostgreSQL, SageMaker), GCP (BigQuery, Dataflow, Cloud Storage), Azure (AKS, Azure ML), Docker, Kubernetes, MLflow, Git, CI/CD

WORK EXPERIENCE

AI Engineer | Northern Trust | USA

Jan 2026 - Present

- Architected an autonomous compliance audit framework using **Python**, **LangGraph**, and **AWS Bedrock** to run multi-agent verification checks across complex institutional fund mandates and SEC disclosures.
- Constructed a high-precision hybrid RAG architecture using **pgvector** on **Amazon Aurora**, **OpenSearch**, and **BGE Embeddings** to query policy benchmarks, reducing compliance audit cycles from 5 days to 45 minutes.
- Built an event-driven serverless ETL pipeline using **AWS Glue**, **AWS Lambda**, **PySpark**, and **Amazon S3** to ingest, clean, and structure unstructured portfolio trade logs and regulatory filings.
- Developed an agentic financial analysis system on **GCP Vertex AI** using **AutoGen** and **Python** to autonomously parse earnings transcripts, extract sentiment signals, and calculate risk metrics.
- Implemented a GraphRAG framework using **Neo4j** and **GCP Vector Search** to map corporate supply-chain dependencies and cross-holdings, improving quantitative factor signal retrieval precision by 34%.
- Integrated Python AI workflows into enterprise accounting systems by engineering **Java** data-processing pipelines and microservices on **Apache Kafka**, scaling execution to process 5 TB of daily market data.

Machine Learning Engineer | Mphasis | India

Nov 2021 - Jul 2024

- Engineered a low-latency streaming pipeline using **Python**, **SQL**, **PySpark**, and **Apache Kafka** to ingest 10M+ daily transactions and process feature engineering pipelines in real time with **Redis**.
- Deployed an ensemble classification system with **XGBoost**, **LightGBM**, and **Scikit-learn**, applying custom loss functions and **NumPy** matrix operations to reduce false-positive rates from 18% to 6.2% while keeping **FastAPI** inference latency under 35ms.
- Developed a multi-modal document extraction framework in **PyTorch** using fine-tuned **LayoutLMv3** and **Transformers** to extract structured fields and redacting PHI/PII data with 99.8% precision.
- Containerized model endpoints using **Docker**, **Kubernetes**, and **REST APIs**, tracking experiment iterations with **MLflow** to deliver a Human-in-the-Loop workflow that raised straight-through processing to 78%.
- Built a multi-horizon forecasting system over 50,000 SKU-store pairs using **Pandas**, **Polars**, and **PyTorch Forecasting**, combining **LightGBM** with **Temporal Fusion Transformers** to capture complex seasonality.
- Conducted exploratory data analysis and statistical optimization using **SciPy** and **Optuna** to improve forecast accuracy (WAPE) by 22.5% and cut annual inventory holding costs by \$3.1M.
- Constructed a Retrieval-Augmented Generation pipeline using **PyTorch**, **Hugging Face**, and **ChromaDB**, integrating dense vector embeddings and **BM25** re-ranking for enterprise document retrieval.
- Fine-tuned open-weights models using **PEFT/LoRA** techniques and deployed low-latency inference endpoints on **Azure Kubernetes Service** using **vLLM**, reducing contract audit turnaround time by 70%.

EDUCATION

Master of Science in Computer Science

Elmhurst University, Elmhurst, IL | Aug 2024 – May 2026

PROJECTS

Digital Twin Multi-Agent AI Platform

- Designed and developed a production-ready multi-agent AI platform using LangGraph, LangChain, and Azure OpenAI to automate career guidance, financial planning, and personalized learning workflows.
- Implemented Retrieval-Augmented Generation (RAG), semantic search, vector databases, and agent orchestration to deliver contextual, multi-step AI responses with improved accuracy and reliability.
- Built scalable RESTful APIs using FastAPI and integrated PostgreSQL for persistent memory, user management, and secure backend services supporting concurrent AI workflows.
- **Technologies:** Python, LangGraph, LangChain, Azure OpenAI, FastAPI, PostgreSQL, React, Docker

AI-Powered 3D Point Cloud Quality Assessment System

- Developed an AI-powered computer vision application for automated quality assessment of construction point cloud datasets using Open3D and geometric analysis techniques.
- Implemented point cloud registration, deviation analysis, and defect detection algorithms to improve inspection accuracy and reduce manual quality verification efforts.
- Built an interactive visualization platform using React and Three.js for real-time 3D rendering, engineering model comparison, and defect analysis.
- **Technologies:** Python, Open3D, Computer Vision, FastAPI, React, Three.js

Brain Age Prediction Using Deep Learning

- Developed a deep learning framework using Convolutional Neural Networks (CNNs) to predict biological brain age from MRI-based neuroimaging datasets.
- Built an end-to-end pipeline including image preprocessing, augmentation, feature extraction, model training, and evaluation to support medical imaging research.
- Published the research in the International Journal of Engineering Research (IJER), demonstrating the application of deep learning techniques in healthcare.
- **Technologies:** Python, TensorFlow, CNN, Deep Learning, Medical Imaging

Customer Risk & Loan Prediction

- Developed supervised machine learning models to predict customer credit risk and loan approval outcomes using structured financial datasets.
- Performed feature engineering, exploratory data analysis, and model optimization to improve prediction performance and support lending decisions.
- Built interactive dashboards using Streamlit and Power BI to visualize customer segmentation, credit risk, and key business insights for decision-makers.
- **Technologies:** Python, Scikit-learn, SQL, Streamlit, Power BI

CERTIFICATIONS

- Google Cloud Certification – Google Cloud
- Microsoft Certified: Azure Fundamentals (AZ-900) – Microsoft
- AI & Data Science Certification – IBM
- Java Programming Certification – Oracle
- Claude 101
- AI Fluency: Framework & Foundations

PUBLICATIONS

AI for Brain Age Prediction Using Deep Learning

International Journal of Engineering Research (IJER)

- Published research on deep learning-based brain age prediction using MRI neuroimaging datasets.
- Developed a Convolutional Neural Network (CNN) framework for image preprocessing, feature extraction, model training, and biological brain age estimation.